# Hieu N. Nguyen

State College, PA

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## Research interests

I'm interested in (i) understanding the fundamental strengths and limitations of learning algorithms through carefully designed, controlled experiments, (ii) developing intelligent systems with capabilities for reasoning, planning, and creativity, and (iii) leveraging these systems to accelerate scientific discovery and tackle real-world challenges at scale.

## Education

**The Pennsylvania State University - University Park** — State College, PA

PhD in Computer Science — Aug. 2025 - Current

- Advisor: Prof. Rui Zhang
- Research Topics: Large Language Models, Reasoning, Mechanistic Interpretability, AI for Scientific Discovery
- Awarded *University Graduate Fellowship*

**VNU - University of Engineering and Technology** — Hanoi, Vietnam

Bachelor of Computer Science - Honors Program — Aug 2022

- Thesis: An application of Neural Tangent Kernel on DEQ Models - Advisors: Dr. Hoang Thanh-Tung and Dr. Ta Viet Cuong.

## Research Experience

**The Pennsylvania State University** — State College, PA

Advisor: Prof. Rui Zhang — Aug 2025 - Current

- Developed a technique to identify critical weights that influence model reasoning behavior. By strategically protecting just 2% of parameters during quantization, the method achieves a 6.57% accuracy improvement on reasoning benchmarks over state-of-the-art compression approaches (ICLR 2026).
- Evaluated the impact of training data characteristics on reasoning styles and coverage loss across SFT and RL post-training, demonstrating critical brittleness in the thinking traces of distilled reasoning models (First-author submission, COLM 2026).

**VinUniversity - College of Engineering & Computer Science** — Hanoi, Vietnam

Research Assistant — June 2023 - Aug 2025

- Developed LLM-SRBench, a rigorous evaluation framework comprising 239 problems designed to minimize memorization and assess true data-driven reasoning in LLMs for scientific equation discovery (co–first author, ICML 2025 Oral). Supervisor: Prof. Khoa Doan.
- Mitigated reward over-optimization in direct alignment algorithms for LLMs by designing and implementing importance sampling techniques, resulting in a publication at NeurIPS 2025. - Supervisor: Prof. Khoa Doan
- Engineered the Correlated Low-rank Structure (CoLR) framework for Federated Recommendation Systems, integrating lightweight low-rank trainable parameters and Homomorphic Encryption to reduce communication payload size by up to 93.75% with only an 8% performance trade-off (first author, WWW 2024 Oral). - Supervisor: Prof. Kok-Seng Wong and Prof. Andrew Le

**FPT AI Center** — Hanoi, Vietnam

AI Resident - Mentor: Dr. Hoang Thanh-Tung — May 2021 - June 2023

- Main research topics: Theoretical foundations of machine learning.
- Formulated a novel cosine similarity-based method for Out-of-Distribution (OOD) detection, improving model robustness and reliability for deployment (Presented at Spurious Correlations, Invariance, and Stability Workshop @ ICML 2023).
- Formulated selective poisoning strategies for clean-label backdoor attacks under a highly constrained, zero-knowledge threat model, maximizing attack success rates using only limited, target-class data, leading to a publication at ICLR 2025.
- Developed an influence function–based approach for labeling error detection. Identified cross-class data as the root cause of IF instability in deep networks and proposed a class-based IF method. This approach substantially improves labeling error detection in NLP tasks without additional computational overhead (ACL 2023).

**VNU-UET** — Hanoi, Vietnam

Advisors: Dr. Hoang Thanh-Tung and Dr. Ta Viet Cuong — May 2021 - June 2022

- Formulated a mathematical proof establishing a linear convergence rate for training monotone Deep Equilibrium (DEQ) models under specific initialization conditions.

## Publications

Why Do Reasoning Models Lose Coverage? The Role of Data and Forks in the Road
 **Nguyen Ngoc-Hieu**, Parshin Shojaee, Minh Phuc Nguyen, Nan Zhang, Chandan K. Reddy, Khoa D Doan, Rui Zhang
 *Submitted to Conference on Language Modeling (COLM)*, 2026

When Reasoning Meets Compression: Understanding the Effects of LLMs Compression on Large Reasoning Models
 Nan Zhang, Eugene Kwek, Yusen Zhang, **Nguyen Ngoc-Hieu**, Prasenjit Mitra, Rui Zhang
 *ICLR*, 2026, URL: `https://openreview.net/forum?id=2za3iNkwXn`

Wicked Oddities: Selectively Poisoning for Effective Clean-Label Backdoor Attacks
 Nguyen Hung-Quang, **Nguyen Ngoc-Hieu**, The-Anh Ta, Thanh Nguyen-Tang, Kok-Seng Wong, Hoang Thanh-Tung, Khoa D Doan
 *ICLR*, 2025, URL: `https://openreview.net/forum?id=1Z3C49JQVf`

Mitigating Reward Over-optimization in Direct Alignment Algorithms with Importance Sampling
Nguyen Minh Phuc, **Nguyen Ngoc-Hieu**, Duy Minh Ho Nguyen, Anji Liu, An Mai, Binh T. Nguyen, Daniel Sonntag, Khoa D Doan
*NeurIPS*, 2025, URL: `https://openreview.net/forum?id=ltPRj2nthL`

LLM-SRBench: A New Benchmark for Scientific Equation Discovery with Large Language Models
Parshin Shojaee*, **Nguyen Ngoc-Hieu***, Kazem Meidani, Amir Barati Farimani, Khoa D Doan, Chandan K. Reddy
***ICML (Oral)***, 2025, URL: `https://openreview.net/forum?id=SyQPiZJVWY`

Towards Efficient Communication and Secure Federated Recommendation System via Low-rank Training
**Nguyen Ngoc-Hieu**, Tuan-Anh Nguyen, Tuan Minh Nguyen, Vu Tien Hoang, Dung D. Le, Kok-Seng Wong
***WWW (Oral)***, 2024, URL: `https://arxiv.org/pdf/2401.03748`

A Cosine Similarity-based Method for Out-of-Distribution Detection
**Nguyen Ngoc-Hieu**, Nguyen Hung-Quang, The-Anh Ta, Thanh Nguyen-Tang, Khoa D Doan, Hoang Thanh-Tung
*Spurious Correlations, Invariance, and Stability @ ICML*, 2023, URL: `https://arxiv.org/pdf/2306.14920`

Class-based Influence Functions for Error Detection
Thang Nguyen-Duc, Hoang Thanh-Tung, Quan Hung Tran, Dang Huu-Tien, **Nguyen Ngoc-Hieu**, Anh T. V. Dau, Nghi D. Q. Bui
*Association for Computational Linguistics (ACL)*, 2023, URL: `https://aclanthology.org/2023.acl-short.104/`

## Skills

**Programming** Python (NumPy, Pandas, Scikit-learn. etc.), C/C++, CUDA.
**ML frameworks** PyTorch, Jax, DeepSpeed, FSDP, verl, Unsloth.
**DevOps** Linux, Shell (Bash/Zsh), Git, Docker, AWS.